Project: Worked examples
Title: Beam and 2D frame analysis
Status: DRAFT · Author: TempestOS (2026-10-23) · Checker: — (—) · Issued: —

1. Scope and method

Linear elastic analysis of a plane frame or continuous beam by the direct stiffness method: reactions, member end forces, axial force, shear, bending moment and deflection along every member with their extremes and positions, for each load combination and as an envelope. Global x to the right, y up; moments and couples anticlockwise positive; sagging moments positive along a member from node i to node j.

WHAT IT CALCULATES	RELATION
Member stiffness, three freedoms per node, Euler–Bernoulli members	axial EA/L and bending EI/L³ terms
End releases (pins) are taken out by static condensation; supports remove freedoms or add spring stiffness.	
Span loads become equivalent nodal loads, the exact fixed-end forces.	
Displacements, by banded LDLᵀ after a reverse Cuthill–McKee ordering (a non-positive pivot is reported as a mechanism)	K u = F
Internal forces and deflections follow from the member end actions and Macaulay functions of the span loads.	
Extremes are found at stations along each member, plus the zero-shear and zero-slope points found by bisection.	

Basis: Plane frame analysis using the stiffness method[1](Ch. 16); Shear, moment, slope and deflection formulas for elastic straight beams[2](Ch. 8, Table 8.1); Combinations of actions (user-entered factors)[3](6.4.3).

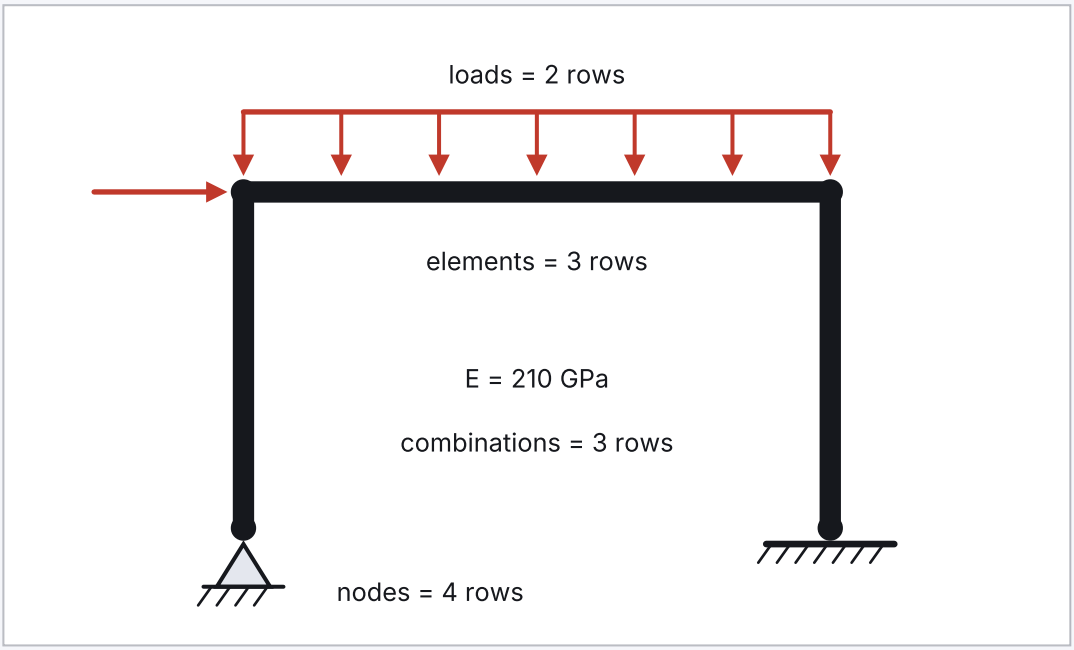


Figure 1 — A plane frame (a portal drawn representatively): the nodes, elements, supports and loads as listed. Not to scale; the labels are the inputs as entered.

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
E	Default modulus of elasticity	210	GPa	spec: BS EN 1993-1-1 3.2.6	1 … 500 GPa
nodes	Nodes	4 row(s)		drawing: DRG-PF-010 rev B	any
elements	Elements	3 row(s)		reference: UC 203×203×46 columns, UB 305×165×40 rafter (Blue Book)	any
loads	Loads	2 row(s)		spec: Load schedule LS-02: roof 15 kN/m; wind 24 kN at eaves	any
combinations	Combinations (blank = each case alone)	3 row(s)		assumption: Characteristic combinations for the worked example.	any
report	Combination to report in detail (blank = first)	G+W		assumption: Gravity plus wind reported.	any
n_{div}	Stations per member	16		assumption: 16 stations per member.	4 … 100

TABLE ROW	VALUES
nodes[1]	id = 1; x = 0 m; y = 0 m; support = pin
nodes[2]	id = 2; x = 0 m; y = 5 m; support = free

TABLE ROW	VALUES
nodes[3]	id = 3; x = 8 m; y = 5 m; support = free
nodes[4]	id = 4; x = 8 m; y = 0 m; support = pin
elements[1]	id = 1; i = 1; j = 2; A = 58.7 cm^2; I = 4568 cm^4; rel_i = no; rel_j = no
elements[2]	id = 2; i = 2; j = 3; A = 51.3 cm^2; I = 8503 cm^4; rel_i = no; rel_j = no
elements[3]	id = 3; i = 3; j = 4; A = 58.7 cm^2; I = 4568 cm^4; rel_i = no; rel_j = no
loads[1]	case = 1; kind = udl; target = 2; a = 0 m; b = 8 m; w1 = -15 kN/m; w2 = 0 kN/m; F_x = 0 kN; F_y = 0 kN; M = 0 kN*m
loads[2]	case = 2; kind = nodal; target = 2; a = 0 m; b = 0 m; w1 = 0 kN/m; w2 = 0 kN/m; F_x = 24 kN; F_y = 0 kN; M = 0 kN*m
combinations[1]	name = G+W; case = 1; factor = 1
combinations[2]	name = G+W; case = 2; factor = 1
combinations[3]	name = G; case = 1; factor = 1

3. Assumptions

A1. Linear elastic, small displacements, first order; no shear deformation, no P–Δ, no buckling. (Euler–Bernoulli direct stiffness.)
A2. Members are prismatic between nodes and connect rigidly unless a release is stated; a release transmits no moment at that end. (Static condensation of the released rotation.)
A3. Distributed loads act perpendicular to the member unless marked as gravity loads, which act in global −y per metre of member length. (Load table convention.)
A4. combinations (Combinations (blank = each case alone)) is taken as 3 row(s). (Characteristic combinations for the worked example.)
A5. report (Combination to report in detail (blank = first)) is taken as G+W. (Gravity plus wind reported.)
A6. n_{div} (Stations per member) is taken as 16. (16 stations per member.)

4. Working

[4] n_{nodes} = nodes
= 4
= 4
Nodes in the model
[5] $n_{elements}$ = elements
= 3
= 3
Members in the model
[6] n_{comb} = combinations
= G+W, G
= 2
Load combinations analysed
[7] comb = reported combination
= G+W
= "G+W"
The combination detailed below
[8] $R1_x$ = reaction at node 1, x
= Σ member end forces − applied nodal loads
= -2.9968 kN
Horizontal reaction at node 1

[9] $R1_y$ = reaction at node 1, y
= Σ member end forces − applied nodal loads
= 45 kN
Vertical reaction at node 1
[10] $R1_m$ = reaction at node 1, moment
= Σ member end moments − applied nodal couple
= -1.4552e-14 kN·m
Moment reaction at node 1
[11] $R1_{x,abs}$ = $|R_x|$ at node 1
= |-2.9968| kN
= 2.9968 kN
Magnitude of the horizontal reaction at node 1
[12] $R1_{m,abs}$ = |M| at node 1
= |-1.4552e-14| kN·m
= 0 kN·m
Magnitude of the moment reaction at node 1
[13] $R4_x$ = reaction at node 4, x
= Σ member end forces − applied nodal loads
= -21.003 kN
Horizontal reaction at node 4
[14] $R4_y$ = reaction at node 4, y
= Σ member end forces − applied nodal loads
= 75 kN
Vertical reaction at node 4
[15] $R4_m$ = reaction at node 4, moment
= Σ member end moments − applied nodal couple
= -2.9104e-14 kN·m
Moment reaction at node 4
[16] $R4_{x,abs}$ = $|R_x|$ at node 4
= |-21.003| kN
= 21.003 kN
Magnitude of the horizontal reaction at node 4
[17] $R4_{m,abs}$ = |M| at node 4
= |-2.9104e-14| kN·m
= 0 kN·m
Magnitude of the moment reaction at node 4
[18] $sum_{R,y}$ = Σ R_y
= all supports
= 120 kN
Total vertical reaction (equilibrium check against the applied load)
[19] $sum_{R,x}$ = Σ R_x
= all supports
= -24 kN
Total horizontal reaction
[20] $M_{max,e1}$ = member 1: max sagging M
= at x = 5 m from node 1
= 14.984 kN·m
Member 1 maximum moment
[21] $M_{min,e1}$ = member 1: min (hogging) M
= at x = 0 m
= 0 kN·m
Member 1 minimum moment

[22] $M_{abs,e1}$ = member 1: max |M|
= max(|14.984|, |0|) kN·m
= 14.984 kN·m
Member 1 greatest moment magnitude (sagging or hogging)
[23] V_{e1} = member 1: max |V|
= 2.9968 kN
= 2.9968 kN
Member 1 maximum shear
[24] N_{e1} = member 1: max |N| (tension +)
= -45 kN
= -45 kN
Member 1 axial force of greatest magnitude
[25] d_{e1} = member 1: max |deflection|
= at x = 5 m
= 74.678 mm
Member 1 maximum transverse deflection
[26] $M_{max,e2}$ = member 2: max sagging M
= at x = 3 m from node 2
= 82.484 kN·m
Member 2 maximum moment
[27] $M_{min,e2}$ = member 2: min (hogging) M
= at x = 8 m
= -105.02 kN·m
Member 2 minimum moment
[28] $M_{abs,e2}$ = member 2: max |M|
= max(|82.484|, |-105.02|) kN·m
= 105.02 kN·m
Member 2 greatest moment magnitude (sagging or hogging)
[29] V_{e2} = member 2: max |V|
= 75 kN
= 75 kN
Member 2 maximum shear
[30] N_{e2} = member 2: max |N| (tension +)
= -21.003 kN
= -21.003 kN
Member 2 axial force of greatest magnitude
[31] d_{e2} = member 2: max |deflection|
= at x = 3.4916 m
= 25.45 mm
Member 2 maximum transverse deflection
[32] $M_{max,e3}$ = member 3: max sagging M
= at x = 5 m from node 3
= 0 kN·m
Member 3 maximum moment
[33] $M_{min,e3}$ = member 3: min (hogging) M
= at x = 0 m
= -105.02 kN·m
Member 3 minimum moment
[34] $M_{abs,e3}$ = member 3: max |M|
= max(|0|, |-105.02|) kN·m
= 105.02 kN·m
Member 3 greatest moment magnitude (sagging or hogging)

[35] V_{e3} = member 3: max |V|
= 21.003 kN
= 21.003 kN
Member 3 maximum shear
[36] N_{e3} = member 3: max |N| (tension +)
= -75 kN
= -75 kN
Member 3 axial force of greatest magnitude
[37] d_{e3} = member 3: max |deflection|
= at x = 0.31515 m
= 75.042 mm
Member 3 maximum transverse deflection
[38] M_{max} = max sagging moment
= member 2 at x = 3 m
= 82.484 kN·m
Maximum moment in the reported combination
[39] x_{Mmax} = position of M_{max}
= from node i of member 2
= 3 m
Position of the maximum moment
[40] M_{min} = min (hogging) moment
= member 3 at x = 0 m
= -105.02 kN·m
Minimum moment in the reported combination
[41] V_{max} = max |V|
= member 2
= 75 kN
Maximum shear in the reported combination
[42] N_{max} = max tension
= member 2
= -21.003 kN
Maximum axial tension
[43] N_{min} = max compression (negative)
= member 3
= -75 kN
Maximum axial compression
[44] d_{max} = max |deflection|
= member 3 at x = 0.31515 m
= 75.042 mm
Maximum transverse deflection
[45] $M_{max,env}$ = envelope max M
= G+W, member 2
= 82.484 kN·m
Maximum moment over all combinations
[46] $M_{min,env}$ = envelope min M
= G+W, member 3
= -105.02 kN·m
Minimum moment over all combinations
[47] $V_{max,env}$ = envelope max |V|
= G+W, member 2
= 75 kN
Maximum shear over all combinations

[48] $d_{max,env}$ = envelope max |δ|
= G+W, member 3
= 75.042 mm
Maximum deflection over all combinations

5. Results

SYMBOL	VALUE	DESCRIPTION
$R1_x$	-2.9968 kN	R_x node 1
$R1_y$	45 kN	R_y node 1
$R1_m$	-1.4552e-14 kN·m	M node 1
$R1_{x,abs}$	2.9968 kN	$\|R_x\|$ node 1
$R1_{m,abs}$	0 kN·m	|M| node 1
$R4_x$	-21.003 kN	R_x node 4
$R4_y$	75 kN	R_y node 4
$R4_m$	-2.9104e-14 kN·m	M node 4
$R4_{x,abs}$	21.003 kN	$\|R_x\|$ node 4
$R4_{m,abs}$	0 kN·m	|M| node 4
$sum_{R,y}$	120 kN	Σ R_y
$sum_{R,x}$	-24 kN	Σ R_x
$M_{max,e1}$	14.984 kN·m	M max, member 1
$M_{min,e1}$	0 kN·m	M min, member 1
$M_{abs,e1}$	14.984 kN·m	|M| max, member 1
V_{e1}	2.9968 kN	|V| max, member 1
N_{e1}	-45 kN	N, member 1
d_{e1}	74.678 mm	δ max, member 1
$M_{max,e2}$	82.484 kN·m	M max, member 2
$M_{min,e2}$	-105.02 kN·m	M min, member 2
$M_{abs,e2}$	105.02 kN·m	|M| max, member 2
V_{e2}	75 kN	|V| max, member 2
N_{e2}	-21.003 kN	N, member 2
d_{e2}	25.45 mm	δ max, member 2
$M_{max,e3}$	0 kN·m	M max, member 3
$M_{min,e3}$	-105.02 kN·m	M min, member 3
$M_{abs,e3}$	105.02 kN·m	|M| max, member 3
V_{e3}	21.003 kN	|V| max, member 3
N_{e3}	-75 kN	N, member 3
d_{e3}	75.042 mm	δ max, member 3
M_{max}	82.484 kN·m	M max

SYMBOL	VALUE	DESCRIPTION
x_{Mmax}	3 m	x of M max
M_{min}	-105.02 kN·m	M min
V_{max}	75 kN	|V| max
N_{max}	-21.003 kN	N max (tension)
N_{min}	-75 kN	N min (compression)
d_{max}	75.042 mm	δ max
$M_{max,env}$	82.484 kN·m	M max, envelope
$M_{min,env}$	-105.02 kN·m	M min, envelope
$V_{max,env}$	75 kN	|V| max, envelope
$d_{max,env}$	75.042 mm	δ max, envelope

6. Checks (governing first)

No pass/fail checks in this calculation.

7. Applicability

No warnings.

8. Verification

```
Template struct.frame-2d@1.3.0 Content hash d545dc5d7d4eb03c Engine calc-engine/1.0.0
Known-answer cases passed 6/6 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0
```

9. References

1. Hibbeler RC. Structural Analysis. 10th ed. Pearson; 2018.
2. Young WC, Budynas RG, Sadegh AM. Roark's Formulas for Stress and Strain. 8th ed. McGraw-Hill; 2012.
3. British Standards Institution. Eurocode: Basis of Structural Design. BS EN 1990. BSI.